INTELLIGENT MARITIME SURVEILLANCE SYSTEM USING YOLO ALGORITHM

A PROJECT REPORT

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in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering

20CS8504 & PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY (AUTONOMOUS)

SAMAYAPURAM – 621 112

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BONAFIDE CERTIFICATE

The work embodies the project report entitled **"INTELLIGENT MARITIME SURVEILLANCE SYSTEM USING YOLO ALGORITHM"** has been carried out by the students LOGA PRIYA S, POOJA P, PRIYADHARSHINI A, RIFFA INFEE R R**.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Maritime surveillance plays an important role in ensuring the safety, security, and efficient monitoring of coastal and ocean environments. With the increase in maritime transportation, illegal fishing, smuggling, and unauthorized vessel activities, there is a growing need for intelligent monitoring systems that can automatically detect and track maritime objects. Traditional surveillance systems mainly rely on radar systems, satellite monitoring, and manual observation through CCTV cameras. These methods often require continuous human supervision and may not provide accurate real-time detection, especially when monitoring large sea areas. The Intelligent Maritime Surveillance System using the YOLO algorithm is designed to improve maritime monitoring by applying advanced computer vision and deep learning techniques. The system processes video feeds captured from surveillance cameras and analyzes each frame to detect objects such as ships, boats, and other marine vessels. By using the YOLO (You Only Look Once) object detection algorithm, the system can identify and classify multiple objects in real time with high speed and accuracy.

Keywords: Maritime Surveillance, YOLO Algorithm, Object Detection, Deep Learning, Computer Vision, Vessel Detection, Security Monitoring.

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SIGNATURE

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
YOLO	-	You Only Look Once
CNN	-	Convolutional Neural Network
AIS	-	Automatic Identification System
RTSP	-	Real Time Streaming Protocol
API	-	Application Programming Interface
ML	-	Machine Learning
UI	-	User Interface

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

The Intelligent Maritime Surveillance System using YOLO Algorithm is designed to monitor maritime environments and detect vessels automatically using computer vision and deep learning techniques. The system uses video feeds from cameras placed in coastal or maritime areas and processes the captured frames to identify objects such as ships and boats. By applying the YOLO object detection algorithm, the system can detect and classify multiple objects in real time with high accuracy. In this project, the video input from surveillance cameras is first captured and processed using image processing techniques. Each frame of the video is analyzed by the YOLO model, which identifies objects present in the scene and draws bounding boxes around them. The detected objects are labelled according to their class, such as ship or boat, allowing authorities to monitor maritime activities effectively. The system reduces the need for continuous human monitoring and improves the efficiency of maritime surveillance. It can help identify suspicious vessel movements, unauthorized entry into restricted areas, or unusual activities in maritime environments. By providing real-time detection and monitoring capabilities, the system enhances maritime security and supports quick decision-making by authorities.

1.1 YOLO ALGORITHM FOR OBJECT DETECTION

YOLO (You Only Look Once) is a popular deep learning algorithm used for real-time object detection. It is designed to detect multiple objects in images or video frames quickly and accurately. Unlike traditional object detection methods that analyze an image in several stages, YOLO processes the entire image in a single pass, which makes it much faster and suitable for real-time applications. The YOLO algorithm works by dividing the input image into a grid of cells. Each grid cell predicts bounding boxes and class probabilities for objects present in that region. A bounding box indicates the location of an

object in the image, while the class probability identifies the type of object detected. By analyzing the whole image at once, YOLO can detect multiple objects simultaneously with high efficiency.

One of the major advantages of YOLO is its speed and accuracy. Since the algorithm processes the entire image in a single network evaluation, it can perform detection in real time. This makes YOLO highly suitable for applications such as video surveillance, traffic monitoring, autonomous vehicles, and maritime monitoring systems. In maritime surveillance systems, the YOLO algorithm is used to detect vessels such as ships, boats, and other marine objects from video feeds captured by cameras. The algorithm analyzes each frame of the video and identifies objects present in the scene. Once detected, the system can highlight these objects using bounding boxes and labels. The use of YOLO in maritime surveillance improves monitoring efficiency and helps detect suspicious or unauthorized vessels quickly. It reduces the need for constant human supervision and provides accurate real-time information to authorities. As a result, YOLO-based systems are widely used in modern intelligent surveillance applications.

1.2 DEEP LEARNING FOR OBJECT DETECTION

Deep learning is a branch of artificial intelligence that focuses on training computers to learn patterns from large amounts of data. It uses neural networks with multiple layers to process complex information such as images, videos, and audio. In recent years, deep learning has become widely used in computer vision applications, especially for object detection and image recognition tasks. Object detection is an important technique in computer vision that involves identifying and locating objects within an image or video frame. Deep learning models such as Convolutional Neural Networks (CNNs) are commonly used for this purpose. These models analyze visual features like shapes, textures, and patterns to detect and classify objects accurately.

In maritime surveillance systems, deep learning-based object detection methods are used to identify vessels such as ships and boats from video streams captured by cameras. Algorithms like YOLO enable real-time detection with high accuracy, making them suitable for monitoring large maritime environments. By using deep learning techniques,

surveillance systems can automatically detect objects and help improve maritime security and safety. Additionally, deep learning models improve their performance by learning from large datasets during the training process. The more diverse and well-labelled the training data, the better the model can recognize objects in different environmental conditions such as fog, waves, low lighting, or long distances. This capability is particularly important in maritime environments where weather and lighting conditions frequently change. By continuously improving the model through training and optimization, deep learning-based surveillance systems can provide more reliable and accurate object detection in real-world maritime monitoring applications.

Several deep learning algorithms have been developed for object detection applications. Some of the well-known models include Region-Based Convolutional Neural Networks (R-CNN), Fast R-CNN, Faster R-CNN, Single Shot Detector (SSD), and YOLO (You Only Look Once). These algorithms differ in their detection strategies and performance characteristics. While some models prioritize accuracy, others focus on achieving real-time detection speed.

Among these models, YOLO has gained significant popularity due to its ability to perform fast and accurate object detection. The YOLO algorithm treats object detection as a single regression problem, meaning that it predicts object locations and class probabilities simultaneously. Instead of analyzing different regions of an image separately, YOLO processes the entire image in one pass through the neural network. This approach significantly improves processing speed, making it suitable for real-time applications such as surveillance systems and autonomous vehicles.

Deep learning-based object detection has been widely used in various fields including traffic monitoring, healthcare imaging, security surveillance, robotics, and maritime monitoring. In maritime environments, deep learning-based object detection plays an important role in monitoring vessel movements and identifying potential threats. Surveillance cameras placed in coastal areas capture video footage of the surrounding sea. Deep learning algorithms analyze this footage and detect vessels such as ships, boats, and other marine objects. The system can track vessel movements and generate alerts when unusual activities are detected.

CHAPTER 2

LITERATURE SURVEY

2.1 AI-BASED MARITIME OBJECT DETECTION USING DEEP LEARNING, A. KUMAR, S. PATEL – 2020

Kumar and Patel (2020) proposed a deep learning-based maritime object detection system designed to identify ships and small vessels from surveillance video streams. Their research addressed the challenges faced in traditional maritime monitoring systems, which often rely on radar or manual observation. The authors implemented a Convolutional Neural Network (CNN) architecture combined with feature extraction techniques to detect vessels in complex maritime environments. The system processed image frames captured from coastal surveillance cameras and applied object detection algorithms to classify vessels into categories such as cargo ships, fishing boats, and patrol vessels.

The researchers evaluated the model using a dataset of maritime images collected from coastal monitoring stations. Experimental results showed that the CNN-based system achieved high detection accuracy even under challenging environmental conditions such as fog, low visibility, and rough sea waves. However, the study identified certain limitations including slower processing speed and difficulty detecting very small or distant vessels. The authors suggested that integrating faster detection algorithms such as YOLO could significantly improve real-time performance in maritime surveillance applications.

Experimental results showed that the CNN-based system achieved high detection accuracy and improved object classification performance compared to traditional image processing methods. The system was able to identify multiple vessels in a single frame while maintaining reliable detection performance under different lighting conditions. However, the researchers reported some limitations in their study. The system required a large amount of labelled training data to achieve high accuracy. Additionally, the processing speed of CNN-based models was relatively slow for real-time surveillance. The authors suggested integrating faster detection models such as YOLO to improve real- time performance.

2.2 AUTOMATIC VESSEL DETECTION USING SATELLITE IMAGES, B. LI, H. ZHANG – 2019

Li and Zhang (2019) presented a vessel detection approach that utilized satellite imagery combined with machine learning techniques to identify maritime objects. The system was developed to monitor large maritime regions where traditional surveillance methods are difficult to implement. The researchers used image segmentation and feature extraction techniques to identify vessel-like structures from satellite images.

The study demonstrated that satellite-based monitoring can cover vast ocean areas and assist authorities in identifying suspicious vessel movements. However, the authors highlighted limitations related to image resolution and processing delays. Satellite imagery often suffers from cloud cover and lower frame frequency, which makes real-time monitoring difficult. The researchers concluded that combining satellite monitoring with camera-based object detection systems could provide a more reliable maritime surveillance solution.

The primary objective of their research was to monitor large maritime areas where ground-based surveillance systems are not feasible. Satellite images were used to identify ships and other marine vessels in open sea environments. The researchers applied image preprocessing techniques such as noise removal and contrast enhancement to improve satellite image quality. Feature extraction methods were then used to identify vessel-like structures within the images. The extracted features were analyzed using machine learning algorithms to classify detected objects.

The study demonstrated that satellite-based monitoring could cover vast maritime regions efficiently. The system was capable of detecting large vessels with reasonable accuracy and provided useful information for maritime traffic monitoring. Despite these advantages, the authors highlighted several limitations. Satellite images often suffer from cloud cover and low resolution, which can affect detection accuracy. Additionally, satellite-based monitoring cannot provide continuous real-time updates. The researchers suggested combining satellite monitoring with camera-based surveillance systems for improved results.

2.3 COMPUTER VISION TECHNIQUES FOR MARITIME SURVEILLANCE, C. RODRIGUEZ, M. SILVA – 2021

Rodriguez and Silva (2021) explored the use of computer vision techniques for maritime surveillance applications. Their research focused on detecting moving vessels in video streams captured by coastal surveillance cameras. The proposed system used image processing algorithms to identify and track maritime objects. The system applied techniques such as background subtraction and motion detection to identify moving objects within video frames. Once an object was detected, classification algorithms were used to determine whether the object was a vessel or another type of moving entity.

The experimental results showed that computer vision-based monitoring systems could significantly improve surveillance efficiency. The system was capable of detecting vessel movement and tracking their positions in real time. However, the study also revealed several challenges. Environmental factors such as waves, reflections, and varying lighting conditions caused false detections in some cases. The authors suggested integrating deep learning algorithms to improve detection accuracy and reduce false alarms.

They investigated the use of computer vision techniques for maritime surveillance applications. Their research focused on applying image processing methods such as background subtraction, motion detection, and object tracking to identify moving vessels in surveillance footage. The proposed system used frame differencing methods to detect moving objects and then applied classification algorithms to identify vessel types. The results showed that computer vision techniques could significantly improve maritime monitoring efficiency. However, the system struggled with environmental disturbances such as wave motion, reflections on water surfaces, and varying lighting conditions. These factors occasionally caused false detections. The authors recommended the use of deep learning-based detection algorithms to improve accuracy and reduce false alarms in real-world maritime environments.

2.4 DEEP LEARNING FOR REAL-TIME OBJECT DETECTION IN SURVEILLANCE SYSTEMS, D. WANG, Y. LIU – 2020

Wang and Liu (2020) investigated the application of deep learning techniques for real-time object detection in surveillance systems. Their research aimed to develop an automated monitoring system capable of detecting multiple objects from video streams. The proposed system used a deep neural network architecture designed for high-speed object detection. The model analyzed video frames and detected objects such as vehicles, people, and maritime vessels. The detection process involved feature extraction, object localization, and classification.

They explored the application of deep learning algorithms in real-time surveillance systems. Their research focused on the use of advanced neural network models for detecting multiple objects in video streams. The authors developed a deep neural network architecture capable of detecting objects such as vehicles, people, and ships with high accuracy.

The system was evaluated using publicly available surveillance datasets, and the results demonstrated that deep learning models significantly outperformed traditional detection methods. Despite the high accuracy achieved, the researchers noted that training deep learning models requires large datasets and significant computational resources. They also emphasized the need for optimization techniques to improve processing speed for real-time surveillance applications.

The experimental evaluation showed that deep learning-based detection models achieved higher accuracy compared to traditional computer vision methods. The system was capable of detecting multiple objects simultaneously while maintaining reliable detection performance. However, the researchers identified certain limitations in their approach. Training deep learning models required large datasets and high computational resources. The authors suggested using optimized algorithms such as YOLO to improve processing speed and enable real-time surveillance applications.

2.5 EFFICIENT MARITIME MONITORING USING CNN-BASED DETECTION MODELS, E. GARCIA, J. LOPEZ – 2018

Garcia and Lopez (2018) proposed a maritime monitoring system based on Convolutional Neural Network detection models. Their research focused on improving vessel detection accuracy in coastal surveillance environments. The system used deep CNN architectures to analyze images captured by surveillance cameras. The model extracted spatial features from the images and used them to identify maritime objects. The dataset used in the study contained images of ships, boats, and other maritime structures.

Garcia and Lopez proposed a CNN-based maritime monitoring system capable of detecting vessels from coastal surveillance cameras. Their study focused on improving detection accuracy by using feature extraction layers that capture spatial characteristics of maritime objects. The model was trained using a dataset containing images of ships, boats, and maritime equipment.

Experimental results showed that CNN-based detection models were capable of identifying vessels with high precision. However, the system faced challenges in detecting partially occluded objects and vessels that appeared very small in the frame. The authors suggested incorporating multi-scale detection techniques to address these issues and improve detection performance in real maritime environments.

The results demonstrated that CNN-based models were effective in detecting vessels with high precision. The system successfully classified different types of maritime vessels and reduced detection errors. However, the system faced difficulties detecting very small vessels that appeared far away from the camera. The authors suggested incorporating multi-scale detection techniques and larger datasets to improve detection performance.

2.6 INTELLIGENT SURVEILLANCE USING YOLO FOR REAL-TIME MONITORING, F. AHMED, K. RAHMAN – 2022

Ahmed and Rahman (2022) proposed an intelligent surveillance framework based on the YOLO object detection algorithm. The study focused on developing a real-time monitoring system capable of detecting objects from live video feeds. The YOLO model was trained to detect multiple object categories simultaneously while maintaining high detection speed.

The researchers demonstrated that YOLO significantly improves detection speed compared to traditional object detection models. The system was able to process video frames in real time, making it suitable for applications such as traffic monitoring and maritime surveillance. However, the authors noted that YOLO sometimes struggles with detecting very small objects in crowded scenes. They suggested further optimization and dataset expansion to improve detection accuracy.

Ahmed and Rahman (2022) proposed an intelligent surveillance framework based on the YOLO object detection algorithm. Their research focused on improving real-time monitoring capabilities in surveillance systems. The YOLO model was trained to detect multiple object categories in video frames. The algorithm divides the input image into grid cells and predicts bounding boxes and object classes simultaneously.

This approach allows the system to perform detection in a single network pass. Experimental results showed that YOLO achieved significantly faster detection speeds compared to traditional detection algorithms. The system was capable of processing video streams in real time while maintaining high detection accuracy. However, the researchers noted that YOLO may struggle to detect very small objects in crowded scenes. They recommended further improvements in training datasets and model optimization to enhance detection performance.

2.7 MARITIME SECURITY MONITORING USING DEEP LEARNING TECHNIQUES, G. BROWN, T. WILSON – 2019

Brown and Wilson (2019) examined the role of deep learning techniques in enhancing maritime security monitoring systems. Their research focused on developing automated surveillance systems capable of detecting illegal maritime activities such as smuggling, piracy, and unauthorized vessel entry. The proposed system used deep neural networks to analyze video streams and detect suspicious activities.

The study concluded that automated surveillance systems could significantly reduce human workload and improve detection efficiency. However, the authors pointed out that system performance can be affected by environmental factors such as weather conditions and camera quality. They emphasized the importance of integrating multiple sensing technologies to improve system reliability.

Brown and Wilson (2019) studied the role of deep learning techniques in maritime security monitoring systems. Their research focused on developing automated surveillance solutions capable of identifying suspicious maritime activities. The proposed system used deep neural networks to analyze surveillance footage captured from coastal monitoring stations. The model was trained to detect vessels and identify abnormal activities such as unauthorized entry into restricted maritime zones.

The experimental results showed that deep learning-based surveillance systems significantly improved detection accuracy and reduced the workload of human operators. The automated monitoring process enabled authorities to detect potential threats more efficiently. Despite these advantages, the system faced challenges related to environmental conditions such as fog and heavy rain. The authors suggested integrating multiple sensor technologies to improve the reliability of maritime surveillance systems.

2.8 OBJECT DETECTION USING YOLO FOR MARITIME APPLICATIONS, H. KIM, S. LEE – 2021

Kim and Lee (2021) investigated the application of the YOLO object detection algorithm in maritime monitoring systems. Their study focused on detecting ships and boats from surveillance video captured in coastal areas. The YOLO model was trained using maritime datasets containing images of various vessel types.

The results demonstrated that YOLO achieved high detection accuracy while maintaining real-time processing speed. The system was capable of detecting multiple vessels simultaneously and tracking their movements. Despite these advantages, the researchers observed that detection performance decreased in extremely poor weather conditions. They recommended integrating image enhancement techniques to improve system performance.

Kim and Lee investigated the application of the YOLO object detection algorithm in maritime environments. Their study focused on detecting vessels from video streams captured in coastal areas. The YOLO model was trained using a maritime dataset containing images of ships, boats, and other marine vessels. The algorithm analyzed video frames and detected objects by drawing bounding boxes around them.

The results showed that YOLO achieved high detection accuracy and fast processing speed, making it suitable for real-time maritime monitoring applications. The system was capable of detecting multiple vessels simultaneously. However, the researchers observed that detection performance decreased under extremely poor weather conditions. They recommended using image enhancement techniques and additional training data to improve detection reliability.

2.9 REAL-TIME VESSEL DETECTION USING DEEP NEURAL NETWORKS, I. FERNANDEZ, L. GOMEZ – 2020

Silva Fernandez and Gomez (2020) presented a deep learning-based framework for detecting maritime vessels in real time using deep neural networks. Their research focused on addressing the challenges faced in traditional maritime surveillance systems, which often rely on radar systems or manual observation. These traditional methods sometimes fail to provide accurate detection, especially in crowded maritime environments or during poor weather conditions. The authors proposed an automated surveillance approach that uses advanced neural network models to analyze video frames captured from coastal monitoring cameras. The goal of the system was to automatically detect vessels and provide real-time monitoring support for maritime security authorities.

The proposed model utilized a deep neural network architecture composed of multiple convolutional layers designed to extract visual features from images. These layers were capable of identifying patterns related to vessel structures such as hull shape, size, and movement patterns. The system first performed image preprocessing to improve image clarity and reduce noise caused by waves, reflections, or lighting variations. After preprocessing, the neural network analyzed each video frame and identified objects that matched the learned vessel features. Once detected, the system highlighted the vessels using bounding boxes and classified them based on their structural characteristics.

The researchers conducted several experiments using a maritime dataset that included images and video frames captured from coastal surveillance systems. The dataset contained different vessel types such as cargo ships, fishing boats, and patrol vessels. The experimental results showed that the deep neural network achieved high detection accuracy for medium and large vessels. The system also demonstrated the ability to detect vessels under different lighting conditions and camera perspectives. This capability made the model suitable for real-time maritime surveillance applications where constant monitoring of vessel movement is required.

2.10 YOLO-BASED SMART MARITIME SURVEILLANCE SYSTEM, J. PARK, K. CHOI – 2022

Park and Choi (2022) developed a smart maritime surveillance system based on the YOLO (You Only Look Once) object detection framework. Their research aimed to improve the efficiency and accuracy of maritime monitoring systems by applying deep learning techniques for real-time object detection. Traditional maritime surveillance systems rely heavily on radar and manual observation, which can be time-consuming and may fail to detect suspicious activities quickly. To address these challenges, the authors proposed an intelligent surveillance system capable of detecting vessels automatically from video streams captured by coastal surveillance cameras.

The proposed system used the YOLO object detection algorithm because of its high speed and ability to detect multiple objects simultaneously. YOLO processes the entire image in a single neural network pass, making it significantly faster than traditional object detection methods. In this system, video frames captured from surveillance cameras were processed by the YOLO model to identify maritime objects such as ships, boats, and other marine vessels. Once detected, the system displayed bounding boxes around the objects and labeled them accordingly. This allowed surveillance operators to monitor maritime activities more efficiently.

The researchers trained the YOLO model using a dataset consisting of maritime images collected from various coastal monitoring stations. The dataset included images of different vessel types operating under various environmental conditions such as daylight, low light, and cloudy weather. Experimental results demonstrated that the YOLO-based system achieved high detection accuracy while maintaining real-time processing speed. The system was capable of detecting multiple vessels simultaneously and tracking their movements across consecutive video frames.

CHAPTER 3

EXISTING SYSTEM

Maritime surveillance is an essential activity for ensuring the safety, security, and proper monitoring of sea environments. Traditionally, maritime surveillance systems rely on technologies such as radar systems, satellite monitoring, Automatic Identification System (AIS), and manual observation using CCTV cameras. These systems are widely used by maritime authorities to track vessel movements, monitor port activities, and detect potential security threats in coastal regions. However, many of these traditional systems have certain limitations that affect their efficiency and accuracy.

One of the most commonly used systems in maritime monitoring is the radar-based surveillance system. Radar systems are capable of detecting large vessels over long distances and are widely used in ports and coastal monitoring stations. These systems transmit radio waves that reflect off objects such as ships and return signals that help determine the location and movement of vessels. Although radar systems are effective in detecting large objects, they often struggle to identify smaller vessels such as fishing boats or small crafts. In addition, radar systems cannot provide detailed visual information about the detected objects, making it difficult to identify vessel types accurately.

Another commonly used maritime monitoring technology is the Automatic Identification System (AIS). AIS is a tracking system used on ships that transmits information such as vessel identity, position, speed, and direction to nearby ships and coastal monitoring stations. This information helps authorities monitor maritime traffic and prevent collisions between vessels. While AIS provides useful data for vessel tracking, it relies on ships to actively transmit their information. Unauthorized or illegal vessels may disable their AIS transmitters to avoid detection, which reduces the effectiveness of the system.

Satellite-based surveillance is also used to monitor large ocean areas. Satellites capture images of maritime environments and transmit them to monitoring centers for analysis. Satellite surveillance is particularly useful for tracking vessels in remote regions where ground-based monitoring systems are not available. However, satellite monitoring has certain limitations such as low image resolution, high operational cost, and delayed data transmission. Satellite images are usually captured at intervals rather than continuously, which makes real-time monitoring difficult.

In addition to radar and satellite systems, many ports and coastal monitoring stations use CCTV cameras to observe maritime activities. These cameras capture video footage of ships entering and leaving port areas. Human operators monitor the camera feeds to identify unusual activities or suspicious vessel movements. While CCTV-based monitoring provides visual information, it requires constant human supervision. Monitoring multiple video streams for long periods can lead to human fatigue, which increases the risk of missing important events.

Another limitation of existing surveillance systems is the difficulty in analyzing large volumes of visual data. Modern surveillance systems generate massive amounts of video footage every day. Manually analyzing this data is time-consuming and inefficient. In many cases, suspicious activities may go unnoticed because human operators cannot continuously monitor all video feeds effectively. This highlights the need for automated systems capable of analyzing video data automatically.

Traditional image processing techniques have been used in some surveillance systems to detect moving objects in video frames. Methods such as background subtraction and motion detection are commonly used for identifying moving vessels. However, these techniques are highly sensitive to environmental conditions such as waves, reflections on water surfaces, and changes in lighting conditions. These factors can lead to false detections or inaccurate results.

Another challenge in existing maritime surveillance systems is the detection of small or distant vessels. Many traditional systems are designed to detect large ships but may fail to identify smaller boats that could be involved in illegal activities such as smuggling or unauthorized fishing. Detecting such vessels requires advanced detection techniques capable of analyzing visual features in complex maritime environments.

In addition, traditional systems lack the ability to automatically classify detected objects. Even if a system detects an object, it may not be able to determine whether the object is a ship, boat, or other marine structure. This limitation reduces the effectiveness of surveillance systems and increases the reliance on human operators for object identification.

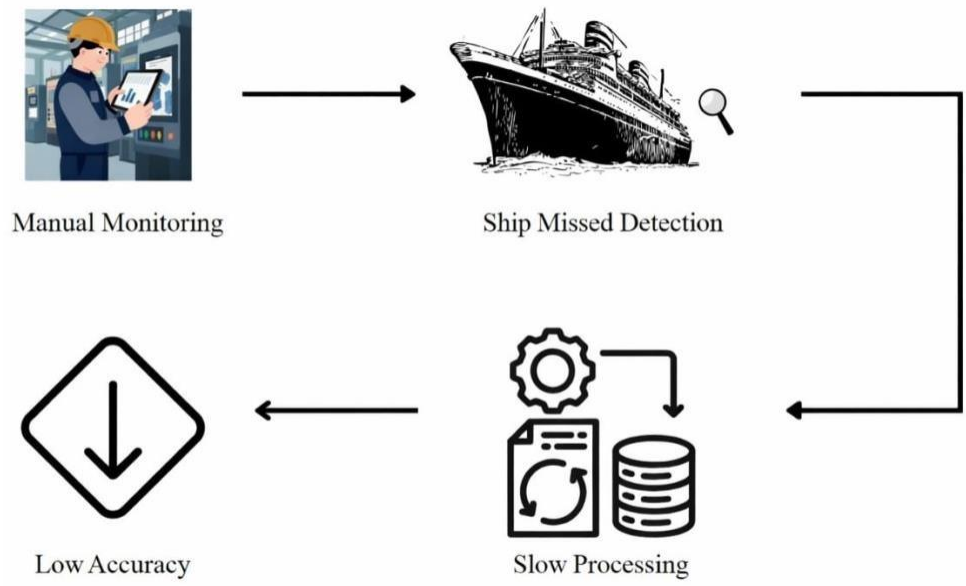

Figure 3.1 Manual Ship Detection Workflow

CHAPTER 4

PROBLEM IDENTIFIED

Maritime environments play a significant role in global transportation, trade, and security. With the increasing number of vessels operating in oceans, ports, and coastal regions, effective monitoring of maritime activities has become very important. Traditional maritime surveillance systems rely on technologies such as radar systems, satellite monitoring, Automatic Identification System (AIS), and manual observation through CCTV cameras.

One of the major problems in existing maritime surveillance systems is the heavy dependence on manual monitoring. Many surveillance systems require human operators to continuously observe video feeds captured by cameras placed in ports or coastal areas. Monitoring multiple camera streams for long periods can lead to human fatigue and reduced attention levels. As a result, suspicious activities such as unauthorized vessel entry, illegal fishing, or smuggling may go unnoticed.

Another important challenge is the difficulty in monitoring large maritime areas. Oceans and coastal regions cover vast geographic areas, making it difficult for traditional monitoring systems to observe every vessel movement effectively. Radar systems can detect large vessels but may fail to identify smaller boats or objects located far away from the monitoring station. Small vessels are often involved in illegal activities, and their detection is crucial for maintaining maritime security.

Traditional surveillance technologies also struggle with identifying and classifying detected objects. Even when a system detects an object, it may not be able to determine whether the object is a cargo ship, fishing boat, patrol vessel, or other maritime structure. Accurate classification is essential for understanding vessel activities and identifying potential threats. Without proper classification, authorities must rely on manual analysis to determine the nature of detected objects.

Environmental conditions present another major challenge in maritime surveillance. Factors such as fog, rain, strong winds, ocean waves, and varying lighting conditions can affect the visibility and quality of surveillance footage. These environmental disturbances can reduce the accuracy of object detection and increase the chances of false alarms. Traditional image processing methods are often unable to handle such complex environmental conditions effectively.

Another problem in existing surveillance systems is the inability to process large volumes of visual data efficiently. Modern surveillance systems generate a huge amount of video data every day. Manually analyzing this data is time-consuming and inefficient. Human operators cannot continuously monitor every video frame, which increases the risk of missing important events. This creates a need for automated systems that can analyze video streams and detect objects automatically.

Existing systems also face limitations in detecting small or distant vessels. Many traditional detection methods focus on large ships that are easily visible on radar or camera systems. These vessels often operate in coastal areas and may engage in illegal activities, making their detection important for maritime security. Another issue in traditional systems is the delay in detecting potential threats. In many cases, suspicious activities are detected only after they have already occurred. The lack of real-time object detection capabilities makes it difficult for authorities to respond quickly to security threats.

Delayed detection can result in serious consequences, including illegal activities going unnoticed or accidents occurring in busy maritime environments. Additionally, traditional surveillance systems require a significant amount of infrastructure and operational cost. Maintaining these systems and employing trained personnel to monitor them continuously also adds to the operational cost. These factors make it difficult for some regions to implement comprehensive maritime monitoring systems.

CHAPTER 5

PROPOSED SYSTEM

The proposed system focuses on developing an Intelligent Maritime Surveillance System using the YOLO algorithm to improve the monitoring and detection of maritime objects. The system is designed to automatically analyze video streams captured from surveillance cameras and detect vessels such as ships and boats in real time. By using advanced computer vision and deep learning techniques, the system can overcome many of the limitations found in traditional maritime surveillance methods.

The YOLO (You Only Look Once) algorithm is chosen for this system because of its ability to perform real-time object detection with high accuracy and speed. Unlike traditional detection algorithms that process images in multiple stages, YOLO analyzes the entire image in a single pass. This allows the system to detect multiple objects simultaneously while maintaining high processing speed. The algorithm divides the input image into grid cells and predicts bounding boxes and object classes for each cell, enabling accurate object detection.

In the proposed surveillance system, the YOLO model is trained using a dataset containing images of different types of maritime vessels. These include cargo ships, fishing boats, patrol vessels, and other marine objects. During the training phase, the model learns to recognize various visual features such as vessel shapes, sizes, and structural patterns. Once the model is trained, it can detect vessels from live video streams captured by surveillance cameras.

The system begins by capturing video input from cameras placed in maritime areas. The video stream is then divided into individual frames, which are processed by the YOLO detection model. The model analyzes each frame and identifies objects that match the learned vessel features. Once a vessel is detected, the system highlights the object by drawing a bounding box around it and labelling it with the appropriate class name.

The proposed system also reduces the need for continuous human monitoring. Instead of relying entirely on human operators to observe surveillance footage, the system automatically analyzes video streams and detects maritime objects in real time. Another important feature of the proposed system is its ability to provide real-time alerts. When suspicious or unauthorized vessels are detected in restricted maritime zones, the system can generate alerts for monitoring authorities.

Overall, the proposed Intelligent Maritime Surveillance System using the YOLO algorithm provides a smart and efficient solution for monitoring maritime environments. By automating the process of vessel detection and reducing dependence on manual observation, the system improves maritime security and enhances real-time monitoring capabilities. The integration of deep learning technology enables accurate object detection and supports authorities in maintaining safer and more secure maritime operations.

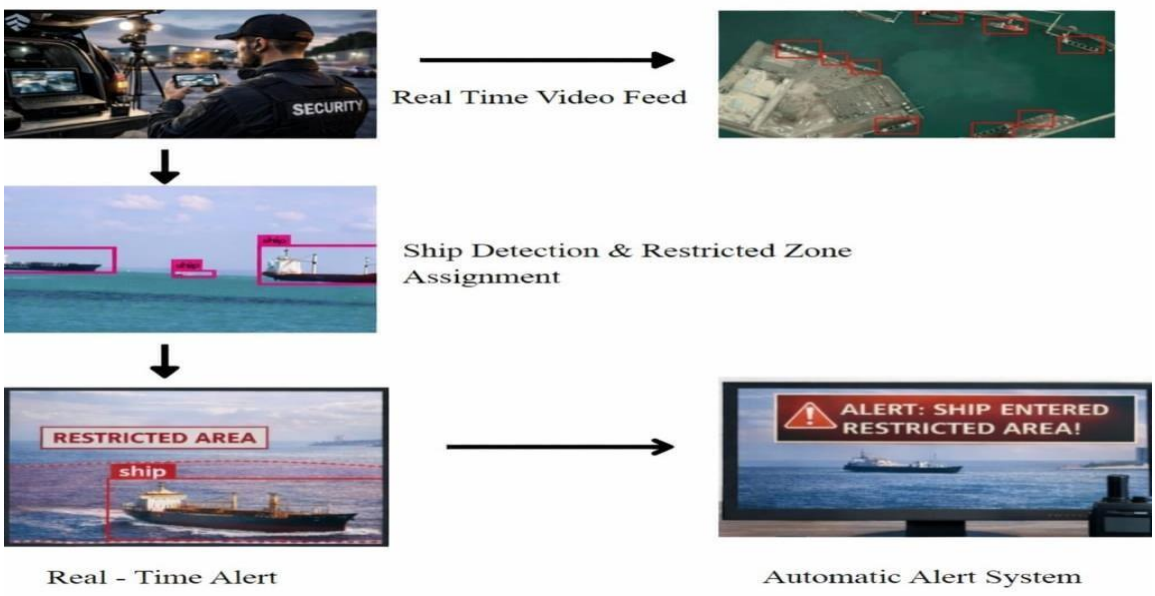

Figure 5.1 Automated Ship Monitoring Architecture

CHAPTER - 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel i5 / AMD Ryzen 5 or above
RAM	Minimum 8 GB
Storage	At least 500 MB free space
Camera	CCTV/ IP Camera/Video feed source

6.2 SOFTWARE REQUIREMENTS

Component	Specification
Operating System	Windows 10 or higher / Ubuntu 18+
Programming languages	Python 3.9+, HTML5, CSS3, JavaScript
Libraries	OpenCV, NumPy, Pandas, PyTorch
IDE	Visual Studio Code / PyCharm / Jupyter

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 LIST OF MODULES

- Video Data Acquisition Module.
- Data Preprocessing Module.
- YOLO Object Detection Module.
- Object Classification and Tracking Module.
- Alert and Monitoring Module

7.2 MODULE DESCRIPTION

7.2.1 Video Data Acquisition Module

The Video Data Acquisition Module is the first and one of the most important components of the Intelligent Maritime Surveillance System. This module is responsible for collecting real-time visual data from maritime environments such as coastal areas, ports, harbors, and maritime borders. The data collected through this module serves as the primary input for the entire surveillance system. Without proper video acquisition, the object detection and monitoring processes cannot function effectively. In this module, surveillance cameras are installed at strategic locations to capture continuous video footage of the surrounding maritime environment. These cameras may include CCTV cameras, high-definition surveillance cameras, or drone-mounted cameras capable of capturing wide-angle views of sea areas. The cameras continuously record video streams that contain information about vessel movements, maritime activities, and other objects present in the monitoring area.

The captured video data is transmitted to the processing system where it is converted into individual frames for further analysis. Each frame contains visual information that can be processed by computer vision algorithms. The video acquisition system ensures

that the captured frames maintain sufficient resolution and clarity so that objects such as ships and boats can be accurately detected by the object detection model.. The collected video streams may be stored in local storage devices or cloud-based servers for future reference and analysis. Efficient data transmission methods are used to ensure that video feeds are delivered to the processing system without delays, enabling real-time surveillance.

7.2.2 Data Preprocessing Module

The Data Preprocessing Module plays a crucial role in preparing the captured video data for object detection. The raw video frames obtained from the Video Data Acquisition Module may contain noise, distortions, or variations in lighting conditions that can affect the performance of the detection algorithm. Therefore, preprocessing techniques are applied to improve the quality of the images before they are passed to the YOLO detection model. In this module, the captured video stream is first divided into individual frames. Each frame represents a still image extracted from the continuous video sequence. These frames are then processed using various image processing techniques to enhance their clarity and consistency. Frame extraction allows the system to analyze each moment of the video separately, which helps in detecting objects accurately.

One of the important preprocessing steps is image resizing. The captured frames may have different resolutions depending on the camera settings. Since deep learning models such as YOLO require images of a fixed size, the frames are resized to match the input dimensions required by the detection model. This step ensures that all images are processed uniformly and efficiently. Filtering techniques are applied to reduce unwanted noise and improve image quality. This helps the detection model focus on meaningful visual features instead of irrelevant disturbances. The preprocessing module may also perform normalization and color adjustments to improve image contrast and brightness. These adjustments make it easier for the detection model to identify objects in varying lighting conditions such as bright sunlight, cloudy weather, or low-light environments.

In addition, the preprocessing module may include data formatting and conversion steps to ensure compatibility with the deep learning model. The processed frames are converted into the required data format that the YOLO model can interpret and analyze. Overall, the Data Preprocessing Module ensures that the input data is clean, consistent, and suitable for object detection. By improving the quality of the captured frames, this module enhances the accuracy and reliability of the YOLO-based detection system. Proper preprocessing significantly contributes to the overall performance of the Intelligent Maritime Surveillance System.

7.2.3 YOLO Object Detection Module

The YOLO Object Detection Module is the core component of the Intelligent Maritime Surveillance System. This module is responsible for detecting and identifying maritime objects such as ships, boats, and other vessels from the processed video frames. It uses the YOLO (You Only Look Once) deep learning algorithm, which is widely known for its high speed and accuracy in real-time object detection applications. The YOLO algorithm works by analyzing the entire image in a single pass through a neural network. Unlike traditional object detection methods that process different parts of an image separately, YOLO divides the image into multiple grid cells and predicts bounding boxes and class probabilities for each grid cell simultaneously. This approach allows the system to detect multiple objects within a single frame quickly and efficiently.

In this module, the preprocessed video frames obtained from the Data Preprocessing Module are provided as input to the YOLO detection model. The model analyzes each frame and identifies objects based on features it learned during the training phase. The training dataset typically contains labelled images of maritime vessels such as cargo ships, fishing boats, patrol vessels, and other marine objects. During training, the model learns to recognize patterns, shapes, and structural features associated with these vessels.

One of the major advantages of the YOLO algorithm is its ability to perform detection in real time. Since the algorithm processes the entire image at once, it significantly reduces detection time compared to traditional methods. This makes YOLO highly suitable for surveillance systems that require continuous monitoring of video streams. The YOLO Object Detection Module can detect multiple vessels simultaneously within a single frame. This capability is especially useful in busy maritime environments where several ships may be present at the same time. By detecting and identifying vessels accurately, the system provides valuable information for monitoring maritime activities.

7.2.4 Object Classification and Tracking Module

The Object Classification and Tracking Module is an important component of the Intelligent Maritime Surveillance System. This module is responsible for identifying the type of detected objects and monitoring their movement across consecutive video frames. After the YOLO Object Detection Module detects objects such as ships or boats, this module further processes the detected results to classify and track the objects in the surveillance area. Object classification refers to the process of categorizing detected objects into predefined classes based on their visual characteristics. In the maritime surveillance system, detected objects may belong to categories such as cargo ships, fishing boats, patrol vessels, or other marine structures. The classification process helps the system understand the type of vessel present in the monitored area, which is useful for maritime traffic monitoring and security analysis.

The classification process uses the information provided by the YOLO detection model. Each detected object is assigned a label based on the learned features from the training dataset. The dataset used for training contains labelled images of different types of maritime vessels. By analyzing features such as shape, size, and structural patterns, the system determines the class of the detected object. Once the objects are classified, the system performs object tracking to monitor their movement across multiple frames of the video. Object tracking is important because vessels are constantly moving in

maritime environments. The tracking mechanism helps the system follow the path of each detected vessel over time. This allows authorities to observe vessel trajectories and analyze movement patterns. The tracking process assigns a unique identifier to each detected object. This identifier allows the system to distinguish between different vessels and track them individually even when multiple objects are present in the surveillance area. By tracking objects across frames, the system can maintain continuous monitoring of vessel movements.

7.2.5 Alert and Monitoring Module

The Alert and Monitoring Module is the final stage of the Intelligent Maritime Surveillance System. This module is responsible for continuously observing the detected objects and generating alerts when unusual or suspicious activities are identified. After the objects are detected, classified, and tracked by the previous modules, the monitoring system analyzes their behavior and movement patterns to ensure maritime safety and security.

In this module, the system monitors the movement of vessels detected in the surveillance area. The tracking information generated by the Object Classification and Tracking Module is used to observe the path, speed, and direction of each vessel. By analyzing this information, the system can determine whether a vessel is behaving normally or entering restricted maritime zones. When the system detects suspicious activities such as unauthorized vessel entry, unusual vessel movement, or a vessel entering a restricted area, it generates an alert. The alert may be displayed on the monitoring screen or sent to the concerned authorities through notification systems. This allows maritime authorities to take immediate action to prevent potential security threats.

The monitoring interface provides real-time visualization of detected vessels. The system displays the video feed along with bounding boxes and labels indicating the detected objects. This helps operators clearly identify the vessels present in the surveillance area. The monitoring dashboard may also display additional information such as vessel location, tracking path, and detection confidence levels.

CHAPTER 8

SYSTEM TESTING

8.1 UNIT TESTING

Unit testing is a software testing method in which individual components or modules of a system are tested separately to ensure that each part functions correctly. The main objective of unit testing is to verify that every module of the system performs its intended task without errors. In the Intelligent Maritime Surveillance System, unit testing is performed for each module such as video acquisition, preprocessing, object detection, classification, and alert generation. In this project, each module is tested independently before integrating it with the rest of the system. For example, the Video Data Acquisition Module is tested to ensure that the surveillance cameras are capturing video streams correctly and transmitting them to the processing system without interruptions. Similarly, the Data Preprocessing Module is tested to verify that the captured frames are properly resized, filtered, and formatted for further analysis.

The YOLO Object Detection Module is tested to ensure that the system can correctly detect maritime objects such as ships and boats from the processed video frames. During testing, sample images and video frames are provided as input, and the system output is verified to confirm that the detection results are accurate. The system should draw bounding boxes around detected objects and assign the correct labels. The Object Classification and Tracking Module is also tested individually to ensure that the detected objects are classified correctly and tracked across multiple frames. The system should assign unique identifiers to each vessel and monitor its movement without losing track. This helps verify that the tracking algorithm works efficiently in real-time conditions. Finally, the Alert and Monitoring Module is tested to confirm that alerts are generated when suspicious activities occur. The system should notify the monitoring authorities whenever unauthorized vessels enter restricted zones or exhibit unusual movement patterns.

8.2 INTEGRATION TESTING

Integration testing is a software testing technique used to verify the interaction between different modules of a system. After unit testing is completed for individual modules, integration testing is performed to ensure that the modules work together correctly as a complete system. The main purpose of integration testing is to identify errors that may occur when different modules communicate and exchange data with each other. In the Intelligent Maritime Surveillance System, integration testing is carried out after all modules such as Video Data Acquisition, Data Preprocessing, YOLO Object Detection, Object Classification and Tracking, and Alert and Monitoring have been tested individually. These modules are then combined and tested as an integrated system to verify that data flows correctly from one module to another.

During integration testing, the video captured from the surveillance cameras is passed to the Data Preprocessing Module, which prepares the frames for further analysis. The processed frames are then sent to the YOLO Object Detection Module to detect maritime objects. Once objects are detected, the results are forwarded to the Object Classification and Tracking Module, which identifies the type of vessel and tracks its movement across frames. Finally, the Alert and Monitoring Module observes the detected objects and generates alerts when suspicious activities occur.

8.3 FUNCTIONAL TESTING

Functional testing is a type of software testing that verifies whether the system performs its intended functions according to the specified requirements. The main objective of functional testing is to ensure that every function of the system works correctly and produces the expected output when valid input is provided. In this type of testing, the system is tested from the user's perspective to confirm that all features operate as designed. In the Intelligent Maritime Surveillance System, functional testing is performed to check whether each function of the system works properly. This includes verifying the system's ability to capture video data from surveillance cameras, preprocess the captured frames, detect maritime objects using the YOLO algorithm,

classify the detected vessels, track their movement, and generate alerts when suspicious activities are identified.

During functional testing, different types of input data such as images and video streams containing ships and boats are provided to the system. The system output is then compared with the expected results. For example, when a vessel appears in the video frame, the system should correctly detect it, draw a bounding box around it, and assign the appropriate label such as ship or boat. If the detected vessel moves across frames, the system should track its movement accurately.

8.4 PERFORMANCE TESTING

Performance testing is a type of software testing used to evaluate the speed, efficiency, and stability of a system under different operating conditions. The main objective of performance testing is to ensure that the system performs well even when processing large amounts of data or operating continuously for long periods. It helps identify performance issues such as slow processing speed, system delays, or high resource usage.

In the Intelligent Maritime Surveillance System, performance testing is conducted to evaluate how efficiently the system processes video streams and detects maritime objects in real time. Since the system continuously receives video input from surveillance cameras, it must be able to analyze each frame quickly without causing delays in detection. The YOLO object detection algorithm is designed to perform high-speed processing, and performance testing helps verify whether the system maintains this speed during operation.

During performance testing, the system is tested with different video inputs and varying numbers of objects present in the frames. The system’s response time, frame processing speed, and detection accuracy are measured. The test ensures that the system can detect multiple vessels simultaneously without affecting the overall performance of the surveillance system.

8.5 USABILITY TESTING

Usability testing is a type of software testing that evaluates how easy and user-friendly a system is for the end users. The main objective of usability testing is to ensure that the system interface is simple to understand, easy to operate, and provides clear information to the users. This testing focuses on user interaction with the system and checks whether users can efficiently perform their tasks without confusion or difficulty. In the Intelligent Maritime Surveillance System, usability testing is conducted to ensure that the monitoring interface is easy for operators and authorities to use. The system should display video feeds clearly and show detected maritime objects with bounding boxes and labels. Users should be able to easily view vessel detection results and understand the information presented by the system.

During usability testing, the system interface is evaluated to check whether the navigation and controls are simple and intuitive. The monitoring dashboard should clearly display the real-time video feed, detected vessels, and alert notifications. Operators should be able to quickly identify ships, boats, and other maritime objects detected by the system without requiring extensive training. Another important aspect of usability testing is evaluating how effectively the system presents alerts and notifications. When suspicious vessel movements or unauthorized entries are detected, the system should generate clear alerts that are easy for the user to understand. The alerts should provide useful information such as the detected object, its location, and the time of detection.

Usability testing also helps determine whether the system reduces the workload of surveillance operators. The system should simplify monitoring tasks by automatically detecting vessels and generating alerts when necessary. This allows operators to focus on important events instead of continuously observing video streams. Overall, usability testing ensures that the Intelligent Maritime Surveillance System is user-friendly and efficient for maritime monitoring authorities. By improving system usability, the system can provide better user experience and support effective decision-making in maritime surveillance operations.

CHAPTER 9

RESULT AND DISCUSSION

The Intelligent Maritime Surveillance System using the YOLO algorithm was developed to detect maritime vessels such as ships and boats from surveillance video streams. The system was implemented using computer vision and deep learning techniques to automatically analyze video frames captured from maritime environments. After implementing the system, several tests were conducted to evaluate its performance in detecting and monitoring maritime objects. During the testing phase, different sets of video inputs containing maritime scenes were used as input to the system. These videos included various vessel types such as cargo ships, fishing boats, and small marine vessels operating in coastal environments. The YOLO object detection model successfully detected the vessels present in the video frames and displayed bounding boxes around them along with the appropriate labels. The detection process was performed in real time, demonstrating the efficiency of the YOLO algorithm for surveillance applications.

The results showed that the system was capable of detecting multiple vessels simultaneously within a single video frame. This capability is particularly important in busy maritime environments where several vessels may be present at the same time. The system successfully identified different vessels and tracked their movement across consecutive frames. The tracking process helped maintain continuous monitoring of vessel activities in the surveillance area. The performance of the detection system was evaluated based on detection accuracy and processing speed. The YOLO algorithm demonstrated high detection accuracy for medium and large vessels that were clearly visible in the video frames. The system was also able to detect vessels moving at different speeds and directions. The real-time detection capability of YOLO allowed the system to analyze video streams efficiently without causing significant delays.

Another important result observed during testing was the system's ability to classify detected objects. Once a vessel was detected, the system labeled it according to the trained classes, such as ship or boat. This classification process helped provide

meaningful information about the detected objects and assisted in understanding maritime activities more clearly.

The object tracking feature of the system also performed effectively. The system assigned unique identifiers to detected vessels and monitored their movement across frames. This allowed authorities to observe vessel trajectories and identify unusual movement patterns. For example, if a vessel changed direction suddenly or entered a restricted area, the system could detect this behavior. The alert generation feature of the system was also evaluated during testing. When suspicious vessel movement or unauthorized entry into restricted zones was detected, the system generated alerts on the monitoring dashboard. These alerts provided timely notifications to surveillance operators, enabling them to respond quickly to potential threats.

Although the system performed well in most conditions, certain limitations were observed during testing. Detection accuracy slightly decreased when vessels appeared very small or were located far away from the camera. Additionally, environmental conditions such as fog, heavy rain, or low lighting sometimes affected the clarity of the video frames, which could reduce detection performance. Despite these limitations, the system demonstrated strong potential for real-world maritime surveillance applications. The integration of the YOLO object detection algorithm significantly improved the speed and accuracy of vessel detection compared to traditional monitoring methods. The system reduced the need for continuous manual monitoring and helped automate the surveillance process.

The results obtained from the testing phase confirm that the Intelligent Maritime Surveillance System can effectively monitor maritime environments and detect vessels in real time. The system provides reliable detection, classification, and tracking of maritime objects, which helps improve maritime safety and security. By automatically analyzing video streams and generating alerts for suspicious activities, the system can assist authorities in maintaining effective surveillance over maritime regions. Overall, the proposed system provides an efficient and intelligent solution for maritime monitoring. The combination of computer vision and deep learning techniques enables the system to analyze large volumes of visual data quickly and accurately.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The intelligent maritime surveillance system using the YOLO algorithm provides an efficient and real-time solution for monitoring maritime environments by accurately detecting and tracking vessels, intrusions, and suspicious activities. With its high-speed object detection capability, YOLO enhances situational awareness, reduces human effort, and improves response time for maritime security operations. The system is scalable, cost-effective, and adaptable to various weather and lighting conditions, making it suitable for ports, coastal areas, and naval applications. Overall, integrating YOLO into maritime surveillance significantly strengthens security, ensures safer navigation, and supports effective decision-making in marine monitoring systems.

10.2 FUTURE ENHANCEMENTS

Future enhancements of the intelligent maritime surveillance system using the YOLO algorithm can focus on improving detection accuracy and adaptability under challenging conditions such as fog, rain, and low-light environments. This can be achieved by integrating advanced versions of YOLO with deep learning techniques like transfer learning and multimodal data fusion using radar, AIS (Automatic Identification System), and thermal imaging.

Another important area of improvement is the integration of predictive analytics and autonomous response mechanisms. By combining YOLO with AI-based behavior analysis, the system can identify suspicious movement patterns, predict potential risks, and generate early warnings. Future systems can also include drone-based surveillance and IoT-enabled smart sensors for wider coverage and flexibility. Furthermore, implementing cloud-based data storage and analytics will support large-scale monitoring and remote accessibility, while strengthening cybersecurity measures will ensure data privacy and system reliability in critical maritime operations.

APPENDIX – A

alert_route :

```
logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(levelname)s] %(name)s: %(message)s"
)
logger = logging.getLogger("main")
app = FastAPI(
    title="Maritime Surveillance System",
    version="1.0.0",
    description="AI Powered Maritime Surveillance Backend"
)
logger.info("==================================")
logger.info("Maritime Surveillance Backend Starting")
logger.info("==================================")
allow_headers=["*"],
logger.info("[SYSTEM] CORS Middleware Enabled")
logger.info("[SYSTEM] Loading API Routes")
app.include_router(camera_routes.router)
logger.info("[ROUTE] Camera Routes Loaded")
```

```
app.include_router(zone_routes.router)
logger.info("[ROUTE] Zone Routes Loaded")
app.include_router(alert_routes.router)
logger.info("[ROUTE] Alert Routes Loaded")
app.include_router(stream_routes.router)
logger.info("[ROUTE] Stream Routes Loaded")
if os.path.exists("alerts"):
    app.mount(
        "/alert-snapshots",
        StaticFiles(directory="alerts"),
        name="alerts-snapshots
    logger.info("[SYSTEM] Static alerts directory mounted at /alert-snapshots")
else:
        logger.warning(app.get("/")
def root():
    logger.info("[API] Root endpoint called")
    return {
        "status": "running",
        "service": "Maritime Surveillance Backend"
    }
@app.get("/health")
def health():
    logger.info("[API] Health check requested")
    return {
        "status": "ok"}
```

```
@app.websocket("/ws")

async def websocket_endpoint(websocket: WebSocket):

    await ws_manager.connect(websocket)

    logger.info("[WebSocket] Client connected")

    try:

        while True:

            # keep connection alive

            data = await websocket.receive_text()

    loggr.info(f"[WebSocket] Received message: {data}")

        except WebSocketDisconnect:

        ws_manager.disconnect(websocket)

        logger.info("[WebSocket] Client disconnected")

    except Exception as e:

        logger.error(f"[WebSocket] Error: {e}")
```

testfile.py :

```
import cv2

# url = "rtsp://wowzaec2demo.streamlock.net/vod/mp4:BigBuckBunny_115k.mov"

url = "http://commondatastorage.googleapis.com/gtv-videos-
bucket/sample/BigBuckBunny.mp4"

cap = cv2.VideoCapture(url)

while True:

    ret, frame = cap.read()

    if not ret:

        print("failed")

        break
```

index.css :

```
:root {
  --bg-primary: #ffffff;
  --bg-secondary: #f4f4f5;
  --text-primary: #18181b;
  --text-secondary: #71717a;
  --accent: #2563eb;
  --border: #e4e4e7;
  --sidebar-width: 240px;
  --header-height: 56px;
}
[data-theme='dark'] {
  --bg-primary: #09090b;
  --bg-secondary: #18181b;
  --text-primary: #fafafa;
  --text-secondary: #a1a1aa;
  --accent: #3b82f6;
  --border: #27272a;
}
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box
```

```
  gap: 0.75rem;
  font-weight: 700;
  font-size: 1.25rem;
  margin-bottom: 1.5rem;
}
.nav-item {
  display: flex;
  align-items: center;
  gap: 0.75rem;
  padding: 0.75rem 1rem;
  border-radius: 0.5rem;
  cursor: pointer;
  transition: all 0.2s;
  color: var(--text-secondary);
  font-weight: 500;
  margin-bottom: 0.5rem;
}
.nav-item:hover,
.nav-item.active {
  background-color: var(--bg-primary);
  color: var(--text-primary);
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.1);
```

```
body {
  font-family: 'Inter', system-ui, -apple-system, sans-serif;
  background-color: var(--bg-primary);
  color: var(--text-primary);
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
}
.dashboard-container {
  display: flex;
  height: 100vh;
  width: 100vw;
}
.sidebar {
  width: var(--sidebar-width);
  background-color: var(--bg-secondary);
  border-right: 1px solid var(--border);
  display: flex;
  flex-direction: column;
  padding: 1.5rem;
}
.sidebar-logo {
  display: flex;
  align-items: center;
```

```
  gap: 0.75rem;
  font-weight: 700;
  font-size: 1.25rem;
  margin-bottom: 1.5rem;
}
.nav-item {
  display: flex;
  align-items: center;
  gap: 0.75rem;
  padding: 0.75rem 1rem;
  border-radius: 0.5rem;
  cursor: pointer;
  transition: all 0.2s;
  color: var(--text-secondary);
  font-weight: 500;
  margin-bottom: 0.5rem;
}
.nav-item:hover,
.nav-item.active {
  background-color: var(--bg-primary);
  color: var(--text-primary);
  box-shadow: 0 1px 3px rgba(0, 0, 0, 0.1);
}
```

```
.main-content {
  flex: 1;
  display: flex;
  flex-direction: column;
  overflow: hidden;
}
.header {
  height: var(--header-height);
  border-bottom: 1px solid var(--border);
  display: flex;
  align-items: center;
  justify-content: space-between;
  padding: 0 2rem;
  background-color: var(--bg-primary);
}
.content-area {
  flex: 1;
  padding: 2rem;
  overflow-y: auto;
  background-color: var(--bg-primary);
```

index.html :

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/vite.svg" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>frontend</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.jsx"></script>
  </body>
</html>
```

APPENDIX – B

SCREENSHOTS

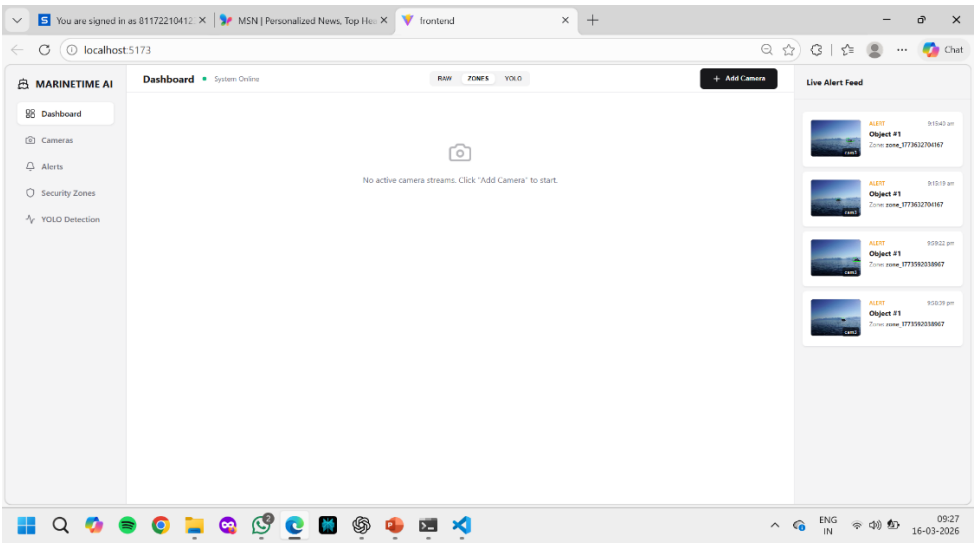

Figure.B.1. Maritime AI Dashboard Interface

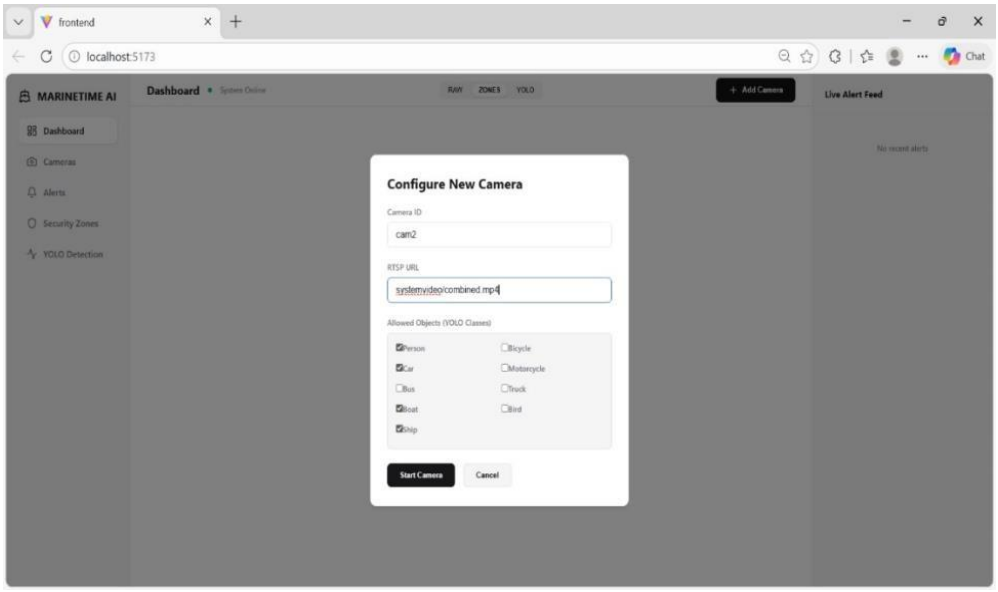

Figure. B.2. Camera Configuration

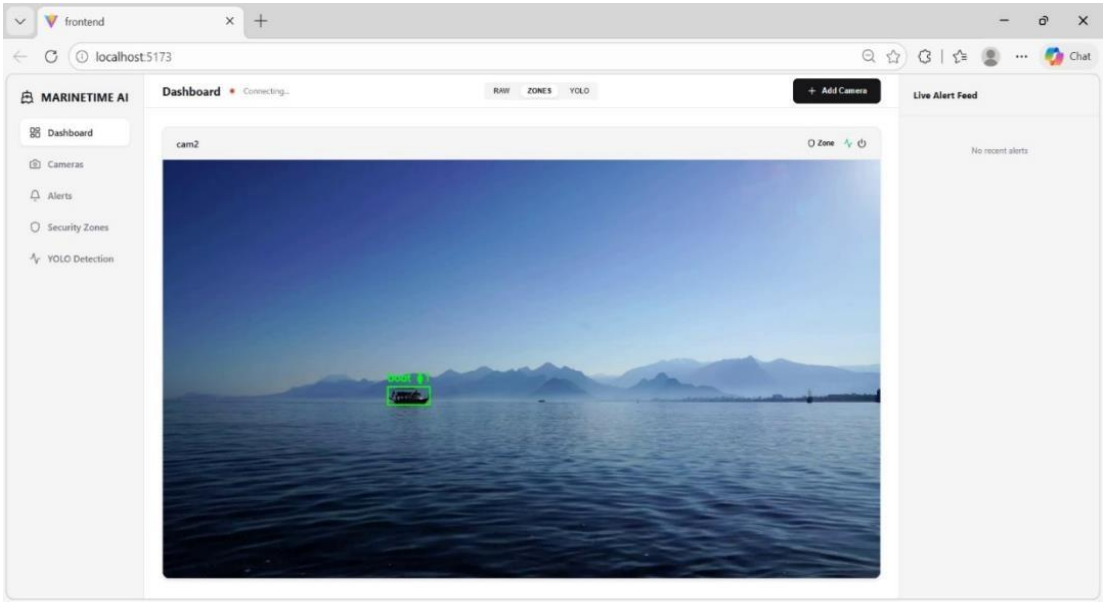

Figure. B.3. Object Detection with YOLOv8

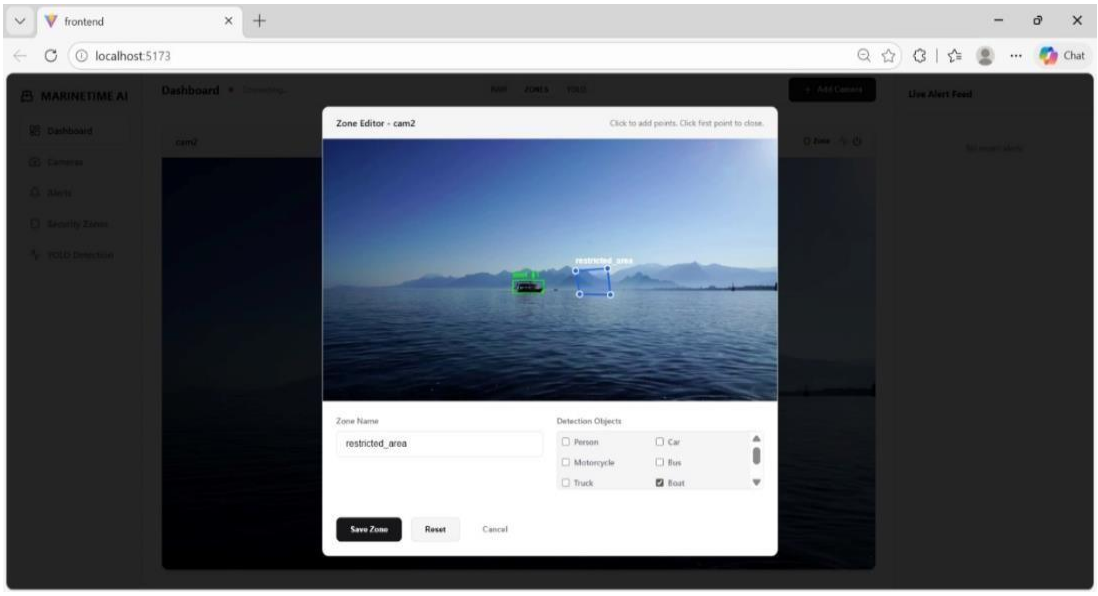

Figure. B.4. Interactive Zone Configuration

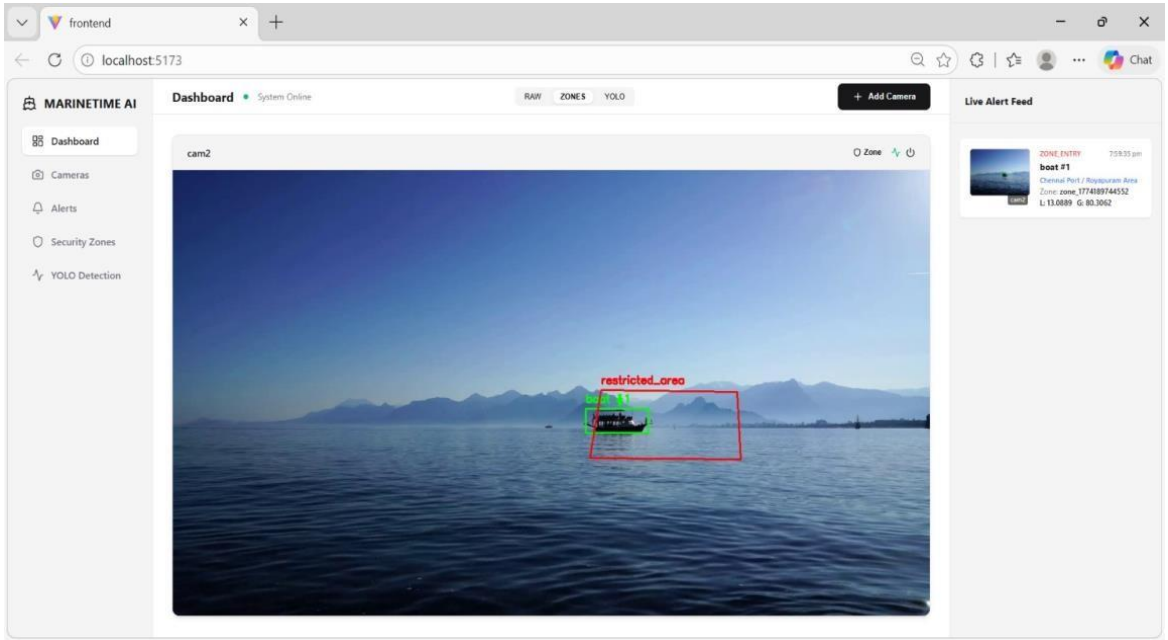

Figure. B.5. Restricted Area Intrusion Alert

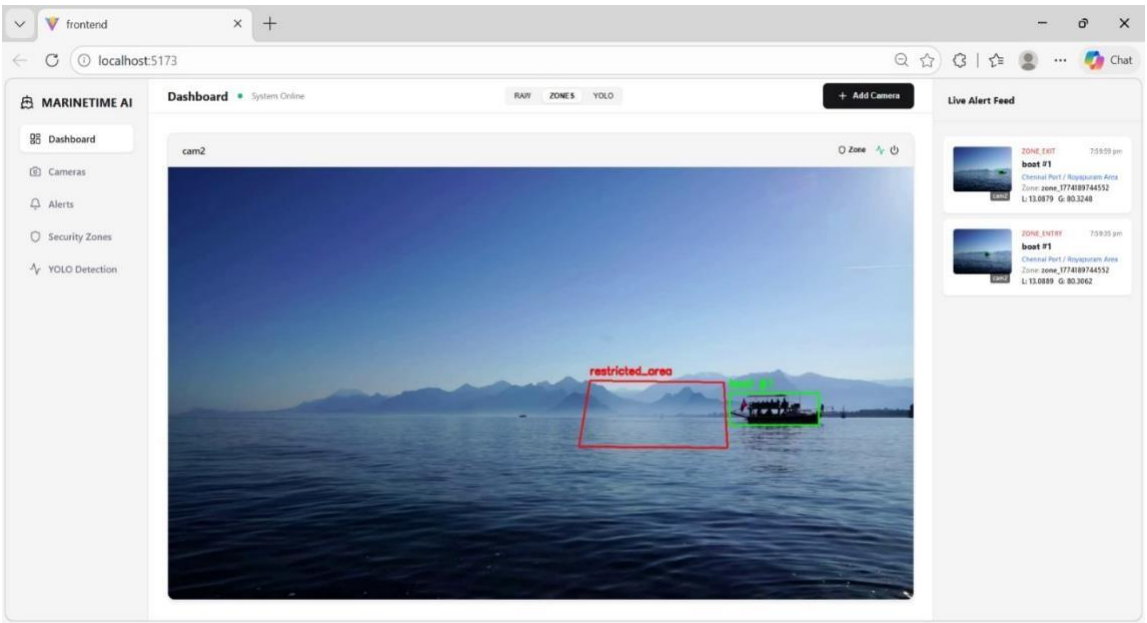

Figure. B.6. Automated Event Logging

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